American International University-Bangladesh (AIUB)

**Department of Computer Science Faculty of Science & Technology (FST)**

# PROJECT TITLE

**AirLink 24/7: A Smart Integrated Airline Passenger Assistance Platform for Real-Time Flight Management, Booking, Security Verification.**

A Software Engineering Project Submitted
By

| **Semester: Spring 25-26** | | **Section: F** |
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## 1. Project Title

**AirLink 24/7: A Smart Integrated Airline Passenger Assistant Platform for Real-Time Flight Management, Booking, Security Verification.**

## 2. Introduction

### 2.1 Overview of the Domain

The aviation industry has transformed global travel by serving millions of passengers through fast and connected digital services. Modern airline platforms require real-time ticket booking, flight scheduling, baggage tracking, passenger verification, and instant communication. Using advanced technologies such as cloud computing and secure verification systems enables the development of scalable and reliable airline solutions. This project focuses on **AirLink24/7**, a unified airline management platform that allows passengers, airline staff, airport officials, travel agents, and security officers to manage operations securely and efficiently 24/7.

### 2.2 Motivation of the Project

Building a full-stack banking system is the perfect way to practice core Software Engineering principles—like system design, testing, and deployment—in a high-stakes, real-world scenario. Many current banking interfaces are clunky or expensive for smaller financial institutions to maintain. This project aims to show that we can build a secure, custom alternative within a single semester while creating a flagship portfolio piece that proves our full-stack skills.

## 3. Background to the Problem

### 3.1 Current System Overview

Many airlines still depend on separate systems for ticket booking, flight tracking, and baggage management. These disconnected platforms often fail to work smoothly together, causing delays in communication and reducing overall operational efficiency. Passengers may experience late flight updates, slow check-ins, and difficulty tracking their luggage in real time.

### 3.2 Limitations of the Existing System

Key limitations include: (a) **Delayed Updates:** Late notifications about flight delays, gate changes, or cancellations.(b) **Poor Baggage Tracking:** No real-time monitoring of passenger luggage.(c) **Slow Check-in Process:** Longer waiting times due to inefficient passenger verification.(d) **Weak Connectivity:** Limited integration with travel agencies and airport systems.(e) **Limited Data Access:** Airlines have restricted access to real-time passenger and operational data.

### 3.3 Need for the Proposed Solution

A unified airline management platform can solve these issues by integrating booking, flight tracking, baggage monitoring, and passenger verification into one secure system. **AirLink 24/7** will provide real-time updates, faster check-ins, better coordination with travel agencies, and improved access to passenger data, creating a smoother experience for both travelers and airline staff.

## 4. Problem Statement

Airlines today don’t have one smart system to handle passenger communication, bookings, security checks, baggage tracking, and performance analytics. Because of using separate systems and manual processes, service is poor, passengers are less satisfied and informed, security risks are noticeable.

## 5. Objectives of the Project

1. Develop a responsive passenger dashboard with flight details, booking history, and travel updates.
2. Implement a secure real-time flight search and ticket booking system.
3. Integrate fast and secure authentication with digital passenger verification.
4. Build a digital check-in and boarding pass generation module.
5. Design a baggage tracking system with real-time luggage status updates.

6. Create a role-based admin dashboard for airline staff, airport officials, and security monitoring.
7. Implement instant notifications for flight delays, gate changes, discounts, and cancellations.

## 6. Solution Overview

We propose a full-stack MERN application (MongoDB, Express.js, React.js, Node.js) built on a three-tier architecture. This includes a React frontend for a smooth user experience, a RESTful API backend, and a MongoDB database for flexible storage.

**Technology Stack**

| Component | Technology | Rationate |
|---|---|---|
| **Frontend** | React.js + Redux | Component-based UI with centralized state for flights, bookings, and passenger data. |
| **Backend** | Node.js / Express.js | Handles high-volume booking requests and real-time flight updates efficiently |
| **Database** | MongoDB / Mongoose | Flexible schema for passengers, flights, baggage, and ticket records. |
| **Authentication** | JWT + bcrypt | Secure login and passenger identity verification with password hashing. |
| **Security** | SSL/TLS + Validation | Protects passenger data and ensures secure communication. |
| **Cloud Storage** | Cloudinary/AWS | Stores passenger documents, digital boarding passes, and baggage images securely. |
| **Notifications** | Firebase / Socket.io | Sends instant alerts for delays, gate changes and booking confirmations |

## 7. Target Users

### 7.1 Primary Users

**Passengers:** Search flights, book tickets, complete check-in, and receive real-time updates.

**Airline Staff:** Manage flight schedules, passenger records, and baggage handling.

### 7.2 Secondary Users

Travel Agencies: Book tickets and manage customer reservations.

Security Officers: Verify passenger identity and ensure compliance.

## 8. Scope of the Project

Several areas can be improved, including the development of a mobile app, the integration of biometric systems, the use of data-driven pricing predictions, and better support for multiple airline alliances.

## 9. Expected Impact and Benefits

Passengers will experience a smoother and more convenient journey with real-time updates and digital boarding passes that simplify every step of travel. At the same time, airlines will gain from centralized dashboards and timely alerts, helping them make better and faster decisions. The system is designed with a strong focus on security and scalability, while also serving as an excellent example of best practices in enterprise software development.

## 10. Conclusion

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This proposal presents **AirLink 24/7**, a secure and scalable airline assistance system designed to unify core airline operations into a single, simple, and user-friendly digital platform. Built with a modern full-stack approach, it aims to replace fragmented airline processes with a centralized system that improves efficiency, reliability, and communication across all stakeholders. The platform enhances the travel experience by providing real-time flight updates, secure booking management, and responsive support services. The scope is realistic for a single-semester timeline, ensuring a functional and well-designed prototype that demonstrates strong software engineering principles, modern system architecture, and effective data management practices.